

Dynamic Guideline: WHO-Based Diphtheria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO clinical guideline identified

- **Guideline title:** *Clinical management of diphtheria: guideline*
- **WHO reference:** WHO/DIPH/Clinical/2024.1
- **Publication date:** 2 February 2024
- **Latest update identified:** 2 February 2024; no newer WHO diphtheria clinical-management guideline was identified up to 29 April 2026. WHO describes this guideline as containing the “most up-to-date recommendations” for clinical management of people with diphtheria [WHO publication page](#), [NCBI Bookshelf](#).
- **Target population:** People with **suspected or confirmed diphtheria**, especially **respiratory diphtheria**.
- **Scope and objectives:** Evidence-based acute clinical management, including **diphtheria antitoxin (DAT)**, **antibiotic therapy**, and supportive care. The guideline explicitly focuses on clinical management and does **not** provide vaccination advice [WHO IRIS PDF](#).

Exactly 5 WHO diphtheria-related guidance documents synthesized

No.	WHO guidance document	Date	Role in this report
1	<i>Clinical management of diphtheria: guideline</i>	2 February 2024	Primary WHO clinical guideline for treatment with DAT and antibiotics WHO publication page
2	<i>Safety and monitoring in patients receiving diphtheria antitoxin</i>	2 February 2024	WHO health-worker guidance on DAT preparation, administration, safety, and monitoring WHO DAT poster page
3	<i>Infection prevention and control and water, sanitation and hygiene measures for diphtheria in health-care settings: operational guide</i>	2024	IPC/WASH outbreak and health-care facility guidance WHO IPC/WASH operational guide page
4	<i>Infection prevention and control measures when caring for patients with suspected or confirmed respiratory diphtheria: a summary</i>	1 March 2024	Practical IPC summary listed by WHO WHO diphtheria health topic page

No.	WHO guidance document	Date	Role in this report
5	<i>Diphtheria outbreaks: comprehensive guidance for public health preparedness and response in the WHO African Region</i>	February 2024	Outbreak preparedness and response guidance, including coordination, case identification, DAT/antibiotics, and catch-up vaccination ReliefWeb / WHO AFRO guidance

Note on “Meaningful youth engagement in a WHO guideline development process”

A WHO evidence brief/webinar titled “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” was identified, scheduled for **11 December 2025**. It concerns the **WHO Abortion care guideline**, not diphtheria [WHO webinar](#). No diphtheria-specific WHO guideline-development case study on meaningful youth engagement was identified in the available evidence. WHO diphtheria materials do, however, emphasize **community engagement**, risk communication, symptom awareness, hygiene messaging, and early care-seeking [WHO EMRO](#).

1.2 Key Recommendations

Clinical recognition and diagnosis

Diphtheria should be treated as a **clinical emergency**. WHO-linked guidance defines suspected respiratory diphtheria as an upper respiratory tract illness—pharyngitis, nasopharyngitis, tonsillitis, or laryngitis—with an **adherent pseudomembrane** of the pharynx, tonsils, larynx, or nose [WHO IPC/WASH operational guide PDF](#). A laboratory-confirmed case requires isolation of *Corynebacterium* spp. by culture and evidence of toxin production [WHO IPC/WASH operational guide PDF](#).

Treatment must not wait for laboratory confirmation when diphtheria is clinically suspected [WHO EMRO](#), [WHO Western Pacific field guide PDF](#).

WHO-based GRADE recommendation table

Recommendation	GRADE Evidence Quality	Strength
Start treatment immediately when respiratory diphtheria is clinically suspected; do not wait for laboratory confirmation.	Low	Strong
Obtain throat and/or nasal swabs for culture and toxigenicity testing, but do not delay DAT or antibiotics.	Low	Strong
For suspected or confirmed diphtheria, use macrolide antibiotics , preferably azithromycin or erythromycin , rather than penicillin antibiotics.	Low	Strong
Administer diphtheria antitoxin promptly when clinically indicated, because DAT neutralizes circulating unbound toxin and delayed treatment increases risk of myocarditis and neuropathy.	Low	Strong

Recommendation	GRADE Evidence Quality	Strength
Do not perform routine sensitivity testing before DAT administration.	Moderate	Strong
Use an escalating DAT dosing regimen based on disease severity and time since symptom onset rather than a fixed dose for all patients.	Very Low	Conditional
Isolate suspected or confirmed respiratory diphtheria cases and implement standard plus transmission-based precautions in health-care settings.	Low	Strong
Monitor closely for airway obstruction, myocarditis, neuritis, renal dysfunction, and shock; provide airway, cardiac, neurologic, and renal supportive care as needed.	Low	Strong
Manage close contacts with risk assessment, prophylactic antibiotics, vaccination-status review, and vaccination if not fully protected.	Low	Strong
After recovery from diphtheria, vaccinate patients who are not up to date, because infection may not confer reliable immunity.	Moderate	Strong
Maintain high population coverage with diphtheria-containing vaccine: 3-dose primary series plus 3 booster doses.	High	Strong

1.3 Guideline Development Methodology

WHO states that the 2024 clinical guideline was developed according to standards for trustworthy guidelines. Recommendations were based on a synthesis of available evidence, with certainty assessed using the **GRADE** approach [WHO IRIS PDF](#).

Evidence review process

- A systematic review was undertaken using a predefined protocol and search strategy.
- Patient-important outcomes were agreed by the Guideline Development Group (GDG).
- Evidence certainty was assessed using GRADE methodology [WHO IRIS PDF](#).
- The guideline was rapidly developed in response to increasing global outbreaks, including outbreaks in Nigeria, Guinea, and neighbouring countries in 2023 [ReliefWeb / WHO](#).

Consensus method

- The GDG reviewed and considered the evidence.
- Outcomes of interest were agreed by the GDG.
- Available summaries do not report detailed voting thresholds or formal dispute-resolution rules [WHO IRIS PDF](#).

Conflict of interest management

- WHO collected written declarations of interest from GDG members.
- The guideline reports **no relevant conflicts of interest**.

- At the first GDG meeting, members were asked to update any declarations verbally; **no verbal conflicts were declared** [WHO IRIS PDF](#).
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1.4 Practical WHO-Based Management Algorithm

Step 1: Suspect diphtheria early

Consider diphtheria in patients with:

- Sore throat, malaise, fever, nasal discharge, hoarseness, cough, dysphagia, or stridor.
- Adherent grey/white pseudomembrane in pharynx, tonsils, larynx, or nose.
- Bull neck, respiratory distress, shock, myocarditis, or neurologic symptoms.

WHO operational material describes early symptoms as malaise, sore throat, and nasal discharge, progressing in severe cases to hoarse voice, cough, painful swallowing, stridor, and shortness of breath [WHO operational protocol PDF](#).

Step 2: Isolate and notify

- Place suspected respiratory diphtheria cases in isolation.
- Use standard and transmission-based precautions.
- Notify public health authorities immediately.
- Begin close-contact identification.

WHO IPC/WASH guidance emphasizes rapid screening, triage, and isolation of suspected or confirmed diphtheria cases in outbreak settings [WHO IPC/WASH operational guide page](#).

Step 3: Collect diagnostic specimens, but do not delay treatment

- Obtain throat and nasal swabs before antibiotics if feasible.
- Request culture, toxigenicity testing, and antimicrobial susceptibility testing where available.
- A suspected case may be discarded if *Corynebacterium* spp. is found but Elek testing is negative, or if PCR is negative for the diphtheria toxin gene [WHO IPC/WASH operational guide PDF](#).

Step 4: Give antibiotics immediately

WHO recommends:

- **Azithromycin or erythromycin** rather than penicillin antibiotics for suspected or confirmed diphtheria [WHO IRIS PDF](#).
- Antibiotic therapy reduces bacterial replication, toxin production, bacterial clearance time, and transmission risk [WHO fact sheet](#).

Step 5: Give DAT promptly when indicated

DAT neutralizes circulating toxin but does not neutralize toxin already bound to tissues [WHO Western Pacific field guide PDF](#). WHO recommends severity- and duration-based dosing:

Clinical presentation	WHO DAT dose
Laryngitis or pharyngitis and symptom duration <48 hours	20,000 IU
Nasopharyngeal disease with extensive pseudomembrane and duration <48 hours	40,000 IU
Diffuse neck swelling, any disease duration ≥48 hours, or severe disease such as respiratory distress or shock	80,000 IU

This dosing approach is a **conditional recommendation** based on **very low-certainty evidence** [WHO IRIS PDF](#).

Step 6: Monitor complications

- Continuous or frequent cardiac assessment for myocarditis.
 - Airway monitoring and early airway support for obstruction.
 - Neurologic monitoring for cranial neuropathy and polyneuropathy.
 - Renal and hemodynamic monitoring in severe disease.
- WHO operational guidance emphasizes supportive therapy for airway compromise and cardiac, neurologic, and renal failure [WHO operational protocol PDF](#).

Step 7: Manage contacts and vaccination

- Close contacts should receive prophylactic antibiotics.
 - Check immunization status and vaccinate if not fully vaccinated [WHO fact sheet](#).
 - People recovering from diphtheria should receive diphtheria-containing vaccine after the acute phase if not up to date [WHO fact sheet](#).
- WHO recommends **six total diphtheria-containing vaccine doses**: a 3-dose primary series beginning as early as 6 weeks of age, followed by 3 booster doses, preferably during the second year of life, at 4–7 years, and at 9–15 years [WHO health topic: Diphtheria](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

- Evidence considered after the WHO clinical guideline release on **2 February 2024** included:
- WHO rapid risk assessments and outbreak documents.
 - ECDC annual epidemiological reports and rapid risk assessments.
 - CDC *Emerging Infectious Diseases* outbreak and antitoxin-procurement reports.
 - Peer-reviewed review evidence on diphtheria antitoxin and antimicrobial resistance.
 - Clinical reference summaries from NCBJ/StatPearls.
 - Clinical-trial evidence related to monoclonal antibody replacement candidates for equine DAT.

Databases and sources searched

- WHO publications, WHO IRIS, WHO health-topic pages.
- ECDC publications.
- CDC *Emerging Infectious Diseases*.
- NCBI Bookshelf / StatPearls.
- PubMed-indexed review evidence available through PMC.
- ClinicalTrials.gov information as reported in CDC evidence summaries.

Inclusion criteria

Included evidence if it addressed at least one of:

- Diphtheria diagnosis or case recognition.
- DAT use, safety, dosing, supply, or alternatives.
- Antibiotic therapy or antimicrobial resistance.
- IPC, outbreak response, contact management, or vaccination.
- Epidemiology after February 2024.

Exclusion criteria

Excluded evidence if:

- It was unrelated to diphtheria.
- It addressed only historical background without clinical or public health relevance.
- It was not from WHO, a major public health agency, a clinical reference source, a peer-reviewed journal, or a clinical-trial registry summary.

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality
WHO Rapid Risk Assessment: <i>Diphtheria, African Region</i> , version 2.0	Regional rapid risk assessment / surveillance synthesis	Rated African regional risk as Moderate and global risk as Low ; most affected countries included Nigeria, Guinea, and Niger; >50% of suspected cases were unvaccinated or had unknown vaccination status WHO RRA PDF .	Low
ECDC Annual Epidemiological Report for 2023, published 2026	Surveillance report	Reported 216 EU/EEA cases in 2023 caused by toxigenic <i>C. diphtheriae</i> or <i>C. ulcerans</i> ; emphasized prompt recognition, laboratory confirmation, treatment, DAT, antibiotics, and contact management ECDC Annual Epidemiological Report 2023 .	Low

Study / source	Design	Key findings	GRADE Quality
ECDC Rapid Risk Assessment 2025: <i>C. diphtheriae</i> ST574 in EU/EEA	Rapid risk assessment / outbreak epidemiology	EU/EEA official case counts declined from 165 in 2023 to 56 in 2024 and 13 in 2025, but ST574 continued circulating in at least five EU/EEA countries and Switzerland ECDC rapid risk assessment 2025 .	Low
CDC <i>Emerging Infectious Diseases</i> : Frankfurt outbreak	Observational outbreak investigation	Described 2023 diphtheria among persons experiencing homelessness in Frankfurt; highlighted need for DAT ideally within 48 hours and noted European DAT supply shortages CDC EID—Frankfurt outbreak .	Very Low
Review: <i>Diphtheria antitoxin treatment: from pioneer to neglected</i>	Narrative/review evidence	Reaffirmed DAT plus antibiotics as core therapy; noted WHO’s macrolide preference is strong but based on low-certainty evidence; highlighted DAT procurement bottlenecks in West Africa 2023–2024 PMC review .	Low
CDC <i>Emerging Infectious Diseases</i> : DAT production and procurement	Policy/practice review with literature and ClinicalTrials.gov search	Identified monoclonal-antibody candidates, including MassBiologics’ S315 and PETA Science Consortium International candidates; cited a 2025 phase 1 S315 study CDC EID—DAT production and procurement .	Low
Sullivan-Bólyai et al., 2025, S315 monoclonal antibody phase 1 study, as cited by CDC EID	Phase 1 healthy-volunteer study	Evaluated safety and pharmacokinetics of human monoclonal antibody S315 against diphtheria toxin; no evidence yet for clinical efficacy in diphtheria patients CDC EID—DAT production and procurement .	Low for safety; Very Low for clinical efficacy
Eurosurveillance 2025 German respiratory diphtheria outbreak	Outbreak report	Reported autochthonous respiratory diphtheria caused by <i>C. diphtheriae</i> in Germany in September 2024 Eurosurveillance .	Very Low

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	GRADE certainty of current evidence	Research priority
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Evidence gap	Why it matters	GRADE certainty of current evidence	Research priority
Optimal DAT dose by severity, age, timing, and disease form	WHO dosing is conditional and based on very low-certainty evidence.	Very Low	Prospective comparative studies or pragmatic trials
DAT availability and alternatives	Global DAT supply remains constrained; delays increase risk of fatal complications.	Low	Stockpile research, procurement systems, monoclonal antibody development
Clinical efficacy of monoclonal antibodies such as S315	Phase 1 evidence addresses safety/pharmacokinetics, not treatment efficacy.	Very Low	Phase 2/3 trials in exposed or infected populations
Comparative antibiotic efficacy	WHO recommendation relies heavily on limited trial evidence and AMR concerns.	Low	Randomized or high-quality observational comparative studies
Antimicrobial resistance surveillance	Resistance to penicillins and macrolides varies by region and time.	Low	Routine susceptibility testing and genomic surveillance
Cutaneous diphtheria DAT indications	DAT is usually not indicated for uncomplicated cutaneous disease, but large membranous ulcers may require individualized assessment.	Very Low	Cohort studies and expert-consensus refinement
Best contact-management strategy in low-resource outbreaks	Contact tracing, prophylaxis, and catch-up vaccination are essential but operationally variable.	Low	Implementation studies in outbreak settings
Youth engagement in diphtheria guideline development	WHO youth-engagement case study identified was for abortion care, not diphtheria.	Very Low	Incorporate adolescents and youth in diphtheria RCCE and guideline-adaptation work

2.4 Updated Recommendations Based on New Evidence

Original WHO-based recommendation	New evidence impact	Updated recommendation as of 29 April 2026
Treat suspected diphtheria immediately; do not wait for laboratory confirmation.	ECDC and outbreak reports reinforce the need for prompt clinical recognition and treatment ECDC Annual Epidemiological Report 2023 .	Unchanged: Strong recommendation. Start DAT and antibiotics immediately when respiratory diphtheria is clinically suspected.
Use DAT promptly when indicated.	CDC and review evidence emphasize that DAT neutralizes unbound toxin and delayed access remains a major problem CDC EID—Frankfurt outbreak , PMC review .	Unchanged, strengthened operationally. Ensure DAT access pathways, stock monitoring, and urgent procurement mechanisms.
Use macrolides—azithromycin or erythromycin—rather than penicillin.	Later reviews confirm WHO’s recommendation but emphasize low-certainty evidence and regional AMR variation PMC review .	Unchanged: Strong recommendation, low-certainty evidence. Prefer azithromycin or erythromycin; adjust if susceptibility data indicate resistance.
Implement isolation and IPC precautions.	ECDC and WHO IPC/WASH materials support rapid isolation, screening, triage, and transmission-based precautions WHO IPC/WASH operational guide page .	Unchanged: Strong recommendation. Isolate suspected/confirmed respiratory cases and protect health workers, caregivers, and visitors.
Conduct contact management and vaccination review.	ECDC reports continued circulation in vulnerable groups; WHO RRA reports many cases in unvaccinated or unknown-status people WHO RRA PDF , ECDC rapid risk assessment 2025 .	Strengthen implementation. Prioritize contacts, migrants, people experiencing homelessness, under-immunized communities, and health/social-care workers.
Maintain high vaccination coverage.	WHO RRA confirms ongoing African Region outbreaks with many unvaccinated or unknown-status cases WHO RRA PDF .	Unchanged: Strong recommendation. Maintain 3-dose primary series plus 3 boosters and deploy catch-up vaccination during outbreaks.
Consider future alternatives to equine DAT.	S315 monoclonal antibody has phase 1 safety/pharmacokinetic evidence but no patient-efficacy evidence yet CDC EID—DAT production and procurement .	No practice change. Monoclonal antibodies remain investigational; equine DAT remains standard of care where indicated.

2.5 Consolidated Dynamic Recommendations

Domain	Current recommendation	GRADE Evidence Quality	Strength
Clinical suspicion	Suspect diphtheria in compatible respiratory illness with adherent pseudomembrane, especially in under-immunized people or outbreak settings.	Low	Strong
Diagnostics	Collect throat/nasal specimens for culture and toxigenicity testing; add PCR and susceptibility testing where available.	Low	Strong
Treatment timing	Do not delay treatment while awaiting laboratory results.	Low	Strong
DAT	Give DAT promptly for suspected/probable respiratory diphtheria when clinically indicated.	Low	Strong
DAT safety	Administer DAT in a monitored setting with readiness to manage hypersensitivity/anaphylaxis.	Moderate	Strong
DAT dosing	Use WHO severity- and timing-based DAT dosing rather than one fixed dose for all.	Very Low	Conditional
Antibiotics	Prefer azithromycin or erythromycin over penicillin for suspected or confirmed diphtheria.	Low	Strong
AMR	Obtain susceptibility testing where possible and strengthen AMR surveillance during outbreaks.	Low	Strong
Supportive care	Monitor airway, cardiac, neurologic, renal, and hemodynamic status; escalate to high-dependency/ICU care when needed.	Low	Strong
IPC	Isolate cases; apply standard and transmission-based precautions.	Low	Strong
Contact management	Identify close contacts, provide prophylactic antibiotics, assess vaccination status, and vaccinate if not up to date.	Low	Strong
Vaccination	Maintain WHO-recommended 6-dose diphtheria-containing vaccine schedule.	High	Strong
Outbreak response	Combine surveillance, laboratory confirmation, DAT/antibiotic access, contact tracing, IPC, RCCE, and catch-up vaccination.	Low	Strong
Monoclonal antibody DAT alternatives	Do not replace equine DAT with monoclonal antibodies outside research or approved emergency frameworks.	Very Low	Conditional

2.6 Bottom Line

As of **29 April 2026**, the WHO **2 February 2024** *Clinical management of diphtheria: guideline* remains the latest identified WHO clinical-management guideline for diphtheria. The central clinical message is unchanged: **recognize diphtheria early, isolate the patient, collect diagnostic specimens, and immediately administer appropriate antibiotics and DAT when indicated—without waiting for laboratory confirmation**. New evidence after the WHO guideline primarily reinforces implementation priorities: ensure DAT access, prefer WHO-recommended macrolides while monitoring antimicrobial resistance, strengthen outbreak surveillance, protect vulnerable under-immunized populations, and maintain high diphtheria-containing vaccine coverage.